

Ian Li

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Education

University of Western Ontario - Honours Computer Science – 3.9/4.0 GPA

2021 - Apr. 2026

Experience

Compiler Engineer Intern - Intel

May 2024 - Oct. 2025

Contributing to **MLIR** ([llvm/llvm-project](https://llvm.org/docs/Project.html)) and Intel MLIR Extensions (IMEX) for **SPIR-V code generation**:

- Wrote MLIR's MathToXeVM pass, converting *fastmath* ops to equivalent native GPU instructions (C++, LLVM, MLIR, SPIRV), **accelerating AI/ML, vector ops on Intel GPUs**
- Added lowerings for MLIR's ArithToSPIRV pass, fixing crashes when converting `arith.index.cast` ops to SPIRV ops

Contributed 100+ PRs to DPC++ SYCL compiler ([intel/llvm](https://github.com/intel/llvm)), enabling GPU programming on **Intel GPUs** (Data Center GPUs (*PVC, Gaudi*), Discrete GPUs (*Arc, Battlemage*), *Aurora Supercomputer*, Iris Xe Integrated GPUs):

- Developed **performance benchmarking** CI (SYCL, Python, Shell, C++), aiding with **performance refactoring** and automatically **detecting regressions** pushed to [intel/llvm](https://github.com/intel/llvm) within 24 hours
- Implemented **GPU-host shared memory prefetch** for DPC++, improving performance by **eliminating latency** from *page faults*
- Debugged critical issues in DPC++ compiler across diverse environments, ranging from deadlocks in distributed PyTorch models (*Intel PyTorch Extension*) to debugging faulty GPU IR behavior (*GenISA, SPIRV, IGC vISA*)

Research Assistant - Ontario Research Centre for Computer Algebra

Sep. 2023 - Apr. 2024

Researched methods for finding range of **CUDA** kernel launch configurations, enabling fast **autotuning** for **GPU compilers** compared to traditional autotuners while also *requiring no manual user input*

- Implemented **LLVM** pass for extracting dimensionality of GPU kernel from **LLVM IR** (via clang), helping to determine problem space of kernel launch parameters for GPU kernel autotuning
- Researched automatic deduction of *valid* kernel configurations (*constraints*) from GPU kernel source code, helping to enable automatic **GPU kernel autotuning** and avoid erroneous kernel configurations when autotuning
- Created static code analysis pass (**Clang Frontend**, Quantifier Elimination via *Maple*) to automatically derive GPU kernel configuration *constraints* from **CUDA** source code

Lead Software Developer - Rentura (Startup)

May 2021 - Feb. 2022

Lead the development and design of a *minimum viable product* (MVP) for a B2C furniture rental platform

- Designed and developed server backend (*Javascript*), with a RESTful API using CRUD endpoints secured with tokens
- Maintained *MongoDB* instances on a **Linux** server, and designed database schemas for an *order management system*
- Developed responsive, mobile-first e-commerce frontend in *React.js* using *Sass* and *Axios* (*MERN* stack)

Projects

Experimental ML Compiler - [ianayl/elohe](https://github.com/ianayl/elohe)

May 2024 - Now

- Writing Python bindings (**pybind11**) for ML/tensor operations that can *JIT-compile* into *optimized GPU kernels*, akin to *PyTorch*
- Devising **MLIR** passes for progressively lowering high-level IR to MLIR GPU dialect to generate/serialize GPU binaries/IR

Experimental Interpreted Language - [ianayl/ex-interpreter](https://github.com/ianayl/ex-interpreter)

Jun. 2020 - Sep. 2021

- Devised an automatic table-based **lexer generator**, and wrote a **recursive descent** parser following LL grammar
- Wrote an interpreter using a **tree-walk evaluator**, and AST passes to generate CFGs/DAG for IR generation

Mindless - [ianayl/mindless](https://github.com/ianayl/mindless)

Hackwestern 2021

Proof-of-concept *unconditionally secure* biometrics password manager that *generates passwords* on-the-fly instead of storing them

- Devised a method to generate consistent passwords using hashing (SHA256) from *facial landmark data* (using **OpenCV**)
- Developed a password manager (**Python, Flask**) that *generates* retrievable passwords using a *face scan*

Shell Site Generator - [ianayl/shsg](https://github.com/ianayl/shsg)

May 2020 - Oct. 2020

Ultra lightweight and portable static site generator that will run on *anything* with a minimal UNIX-like environment

- Developing a static site generator using POSIX shell and PCRE Regex: *No dependencies other than a UNIX shell & coreutils*
- Created a transpiler (Shell) for transpiling markdown to themed webpages with no alternative runtimes required

Technical Skills

Programming Languages	C, C++, Python, CUDA, SYCL (DPC++), Bash / POSIX Shell (+ a lot more, feel free to ask!)
Tooling	LLVM, MLIR, Linux, Clang (Frontend), Github Actions, GDB/LLDB, Nix/NixOS
Interests	(GP)GPU compilers, AI/ML compilers and accelerators, Heterogenous Computing